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Insightful Clinical Assistance for Anemia Prediction with Data Analysis and Explainable AI

Kasthuri.E[a], S.Subbulakshmi[b], Rajasree Sreedharan[c]

[a]Research Scholar, Department of Computer Science and Engineering
[b]Assistant Professor, Department of Computer Science and Applications, Amrita School of Computing, Amritapuri, India
[c]Associate Professor, Division of Nephrology, Department of Pediatrics, Medical College of Wisconsin, Milwaukee, Wisconsin, USA
kasthurie@am.amrita.edu

Abstract

Anemia among young people has positive correlation with susceptibility and impede cognitive development.As per WHO 2019 statistics global prevalence rate of anemia is 33 percent. This work aims to predict anemia with inferences to identify the root cause of anemia. Prediction is done with machine learning algorithms with proper pre-processing, data normalization, and class imbalance approaches. Explainable AI, a transparent model, is used to enhance the prediction which enables to draw inferences. LGM Boost learning model prediction has a 91% accuracy and explainable AI framework predictions of most contributing attribute is accurate when compared to other models. This approach helps healthcare professionals make informed decisions on the most optimal treatment for anemia.

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Keywords: Explainable Artificial Intelligence; Machine learning; Anemia; Data preprocessing.

1. Introduction

Despite the considerable progress made in economic, healthcare, and social development on a worldwide scale, the issue of anemia continues to persist prominently among women and children. Anemia mostly occurs due to a reduction in the number of erythrocytes inside the bloodstream [1]. Based on the global statistics provided by the World Health Organization for the year 2019, it is seen that the prevalence of anemia among women in the age group of 15-49 years is around 29.9 percent. Anemia may lead to fatigue, short breath, conjunctivitis[2], mucous membranes, cold extremities due to reduction of oxygen in the blood or poor circulation of blood, dizziness, cognitive impairment, weakened immune system, rapid heartbeat, heart failures due to extra strain to heart, pre-term delivery and weight reduction among childbirth. The United Nations 2030 goal for Sustainable Development has included the objective of increasing the incidence of anemia among women aged 15-49. Anemia affects a significant proportion of

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the worldwide population, with about one-third 33 percent being affected [3]. The accuracy was impacted due to the use of a limited dataset that did not undergo data pretreatment processes. The study uses machine learning and deep learning models to detect anemia using photos of conjunctiva, fingernails, and palms. Deep learning models have shown greater prediction ability, but their ability to provide insights into the fundamental mechanisms contributing to anemia is limited. Twitter analysis is used to predict anemia by identifying terms related to it and finding the strength of those links. However, most people affected by anemia fall into economically backward categories. Most research has provided more accurate predictions about the existence of anemia among individuals using varied sets of data. Prediction would be useful if it combines the inference about the reason for the prevalence of anemia, as it would act as a clinical aid for taking preventive actions to overcome this disease. The proposed research aims to create a different prediction model enabled with Explainable AI that provides insights into the features that influence anemia prediction, helping practitioners assess the reason for anemia and provide better treatment. The model is created using a publically available tabular data set with nearly 15300 records collected over a period of 5 years. A comparative study is conducted to assess the model's accuracy and the impact of various pre-processing techniques on the effectiveness of machine learning algorithms. A meticulous process is devised to assess performance, which may be applied to other diagnostic models.

This article is ordered as follows. Chapter 2 Literature survey, describes existing work; Chapter 3, elaborates Methodology with pre-processing, machine learning models, and explainable AI used in this work. Chapter 4 investigates Evaluation Metrics, while Chapter 5 provides an overview of the contributions, The work performed in developing this anemia disease diagnosis system is outlined in Chapter 6, and Chapter 7 concludes the study on the identification system for anemia and suggests potential areas for future improvement.

2. Literature Survey

Wemyss, T. A et.al., [9] developed a predictive technique for the identification of anemia in children and newborns residing in Ghana. The objective of this study is to examine and assess several techniques for modifying the surrounding illumination and assessing the degree of redness in images. The study primarily concentrated on a specific area of interest and aimed to conduct colorimetric measurements from two distinct regions of interest, namely the lower eyelid and the lower lip. The provided sample did not consist of a substantial percentage of persons with severe anemia. As a result, the accuracy of predicting the severity of anemia was not improved. Research elaborated in [8] aims to predict anemia among pregnant women in Ethiopia by using ensemble machine-learning algorithms. The Ethiopian population health survey data underwent appropriate preprocessing techniques to improve the data quality provided to the prediction model. The prediction model was constructed using decision trees, random forests, cat boost, and severe gradient boosting techniques, both with and without class decomposition. In the study [4], researchers investigated the relationship between symptoms of illness and the emotions expressed by patients on the Twitter network. Although there has been accurately predicting anemia, the widespread adoption of these methods across a broad consumer base has been unsuccessful. Iron deficiency is a prominent etiological factor contributing to the development of anemia in humans. Kwak et al.[5] conducted a comprehensive investigation to illustrate the relationship between iron deficiency and the incidence of seizures in children. The meta-analysis findings indicated a moderate correlation between the variables under investigation. However, the experimental study conducted was limited by many constraints, such as the exclusion of randomized controlled trials from the analysis. ThalPred [7] was identified in a cohort of Thai people using a retrospective analysis of laboratory data. A discriminant model was constructed to evaluate the presence of thalassemia and anemia diseases, using five distinct machine learning techniques: k-nearest neighbour, decision tree, random forest, artificial neural network, and support vector machine. The evaluation and comparison of performance were conducted to thirteen pre-existing discriminant equations and indicators A novel methodology was developed for the non-invasive diagnosis of anemia, using a combined approach strategy that integrates modified Kalman filtering (KF) with penalty regression. The Kalman Filter (KF) has undergone modifications to accommodate temporal variations in medical data. The use of the suggested Kalman Filter (KF) [6] has the potential to decrease the variance of the dataset, hence potentially facilitating the subsequent task of pattern identification. After conducting a comprehensive analysis of existing research studies on the prediction of anemia, using diverse approaches and datasets, we have determined that our study would specifically concentrate on the

accurate prediction of anemia via the use of explainable artificial intelligence methods. Previous studies successfully predicted the presence or absence of anemia in individuals, but they were unable to ascertain the underlying cause of anemia. Work done by Yagin FH, Cicek ˙IB reveals the importance of Explainable AI for identifying COVID gene biomarkers[**?**]. With reference to the above research this study aims to provide clinical support in the identification of the underlying etiology of anemia, enabling the administration of appropriate therapeutic interventions for its resolution

3. Methodology

Anemia is a prevalent disease among children and women worldwide, with a 40%-52% chance of further seizures. Epilepsy is one of the diseases most common among kids and it is a disorder where unprovoked seizures frequently happen. Iron deficiency anemia is linked to more seizures in some groups[10]. An efficient anemia identification system is essential for improving society's healthcare system. The research aims to perform data analysis, create models for anemia predictions, and generate inferences about the main causes using explainable AI. This will help identify treatment processes for individuals with anemia and improve the overall healthcare system.

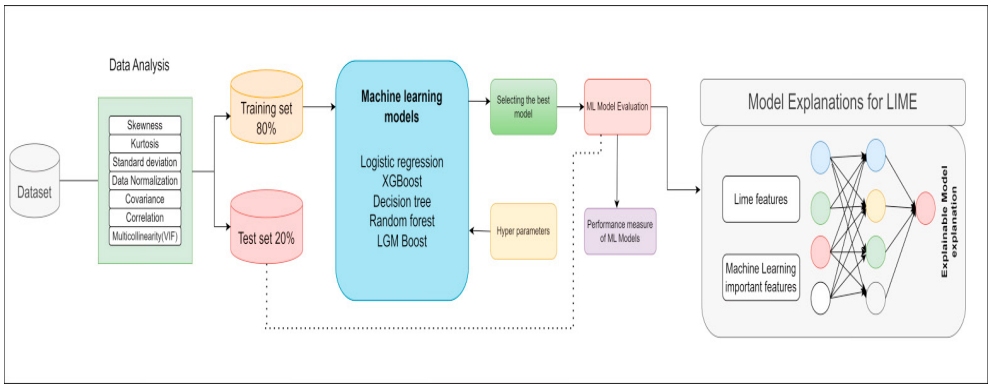


Fig. 1. The architecture of an Anemia Prediction System using Machine Learning Models and Explainable AI

3.1. Dataset description

The publicly accessible dataset which is created with the data obtained from Faculty of Medicine, Tokat Gaziosmanpaşa University, Turkey related to anemia is used in this research work. It contains whole blood count test results gathered from 15,300 individuals from 2013-2018 and it reveals that 1,019 (7%) of the patients had HGB-anemia, 4,182 (27%) went through iron insufficiency, 199 (1%) encountered B12 deficiency, and 153 (1%) was having folate deficit, while the majority, 9,747 (64%) did not have any anemia.

3.2. Data preprocessing and analysis

3.2.1. Outlier removal

Datapoints in a dataset that significantly deviate from other observations [11, 12] are called outliers. The acceptable data range is between Q_2 -1.5 * IQR(lower limit)and Q_3 - 1.5 * IQR(upper limit) , where IQR is calculated using the formula given below. Points beyond this range which are considered to be outliers are dropped from the dataset.

$$IQR = Q_3 - Q_4 \tag{1}$$

3.2.2. Univariate analysis

Data Normalization: Normalization is a technique that transforms attributes to have a mean value of 0 and a standard deviation of 1, improving their effectiveness when the attribute distribution follows a Gaussian distribution[13]. It reduces bias, improves numerical stability, speeds up training, and simplifies feature comparisons. Each numeric attributes in the dataset is normalised using the Z-score method, with the following equation:

$$Y_{\text{new}} = \left(\frac{Y - Y_{\text{mean}}}{S\,tandardDeviation} \right) \tag{2}$$

Skewness: Distribution curves are used to assess skewness in a dataset, a term referring to the asymmetry in the data distribution curve when inclined towards either side. Skewness can affect model assumptions and performance. It helps understand the spread of the data-points in the dataset[14]. Skewness in dataset should be corrected using cube root transformation in order to produce normalized data. Skewness is calculated using a formula.

$$S\,kewness = 3 \left(\frac{mean - median}{S\,tandardDeviation} \right) \tag{3}$$

Kurtosis: It is used to illustrate the nature of data distribution in the data set. It represents the degree to which a distribution tails are heavier than those of a normal distribution[15]. The formula for calculating the kurtosis is given below where, represents an ith variable, mean of the distribution, n total number of variables in the distribution

$$Kurtosis = n * \left(\frac{\Sigma_i^n (z_i - \bar{z})^4}{\Sigma_i^n (z_i - \bar{z}^2)^2} \right) \tag{4}$$

3.2.3. Bivariate analysis

Covariance: Variables can show a linear association [20], which can be explained by examining the covariance between two variables and it does not reveal the variable's relationship strength. When values are not centrally located, they are calculated by the average of the product of each sample's values. The computation is determined using the below formula, in which A and B are variables.

$$Covariance(A,B) = S\,um(A - mean(a) * (B - mean(b)) * \frac{1}{n-1} \tag{5}$$

Correlation: It helps to assess the strength of relationship between the two variables. Positive correlations occur when two variables pass through in the same direction. A negative correlation occurs when one variable rises and others decline [19]. No correlation or association means no relationship between variables. To calculate, use the formula below, where m and n are variables.

$$Correlation = \frac{cov(m,n)}{\sigma_m * \sigma_n} \tag{6}$$

Multicollinearity: Multicollinearity is significant when the variance inflation factor (VIF) for variables is relatively large. It gives a quick approximation of how much a variable adds to the standard error of categorization. There are several methods for reducing or combining collinear variables to address the multicollinearity problem. In anemia dataset, we found multicollinearity and removed it with feature engineering. The formula for calculating VIF is

$$VIF = \frac{1}{1 - R_i^2} \tag{7}$$

3.3. Machine Learning Models

3.3.1. Logistic Regression

It is a classification method designed to determine the relationship between inputs and the probability of a specific result. When diagnosing different medical disorders based on specific symptoms and traits, it is highly helpful [17]. The Logistic Regression is a kind of predictive analysis, just like other regression analysis. It determines the expected likelihood of the result. By including the logit function, the influence of the outlier in a logistic regression is lessened.

3.3.2. XGBoost

XGBoost, also known as Extreme Gradient Boosting, is a machine-learning technique used to create gradient-boosting decision trees (GBDT). It effectively manages sparsity patterns and tackles overfitting in complex models. The methodology uses LASSO and Ridge regularization approaches and features built-in cross-validation capabilities for selecting the most suitable number of boosting iterations. The distributed weighted quantile sketch approach is used to determine the optimal number of split points among weighted datasets.

3.3.3. Decision tree

The hierarchical tree-structured approach is utilized to extract significant results from diverse inputs, contributing significantly to decision-making processes[18]. The model construction process identifies hidden patterns within the input data, determining information gain, index, and dividend growth rate based on optimal partitioning of nodes. The dataset is divided based on the parameters demonstrating the lowest entropy or highest gain in data, ensuring accurate and efficient data analysis.

3.3.4. Random Forest

It is a meta-estimator fitting several decision tree classifiers on various dataset subsamples and uses averaging to improve prediction accuracy and control over-fitting. A random forest with a specific number of trees is created with any supervised classification techniques. The classifier won't overfit the model if the forest has more trees. The ensemble learning technique classifies outcomes based on individual trees and uses many decision trees to train. The random forest reduces the likelihood of overfitting by using many trees.

3.3.5. LGM Boost

The algorithm discussed is an enhanced iteration within the Gradient Boosting Decision Tree (GBDT) framework, utilizing histogram-based approaches. It divides continuous eigenvalues into smaller intervals and creates a histogram, identifying the most suitable points for feature segmentation. This technique reduces computational time and memory consumption[21], making it promising for real-time anemia prediction due to its strong theoretical foundations in data feature extraction and parallel processing, the LightGBM method has promise for real-time anemia prediction.

3.4. Explainable AI

Explainable artificial intelligence is increasingly important in the healthcare sector to make machine learning models less opaque[22, 23]. There are two main methods to explain AI models: model-specific, which is limited to specific versions, and model-agnostic, which applies to all types. Early AI adoption focused on building human-understandable models, rather than providing operational descriptions, and now our aim main objective is to make ML models more interpretable.

3.4.1. Locally interpretable model agnostic explanations (LIME)

It is a white box model providing interpretations to clearly explain individual predictions. LIME acts locally which is observation-specific and provides explanations for the prediction about each observation[24]. It aims to fit a local model using sample data points that are comparable to the observation being explained. This can fall under the umbrella of interpretable models, which also cover linear models, decision trees, etc. Feature selection can be done using explainable AI for increasing accuracy. LIME explanations for each observation c is calculated as follows:

$$\emptyset(c) = argminB_{b\in B}\,(L(a, b, \pi d) + \Omega(b)) \tag{8}$$

where B is the class of potentially interpretable models, $b \epsilon B$: a statement that is considered as a model.π_d(e): Proximity measure of an instance e from D. $\Omega_b(b)$: a measure of the explanation's complexity $b \epsilon B$. Such an explanation helps to interpret the results and aids in making appropriate decisions relevant to the specific situations.

4. Evaluation Metrics

The evaluation of the efficacy of various machine learning models is conducted via the assessment of their accuracy, precision, recall, and F1-score. All the above evaluation scores are retrieved from the confusion matrix generated for every model separately. It takes into consideration classification results of both training and test dataset. It considers correct and wrong predictions made by the models. Accuracy is a metric that quantifies the proportion of correctly classified samples to the total number of samples inside a specified model. Precision is a quantitative measure that evaluates the proportion of correctly detected positive cases among all positive predictions made by various models. The classification of recall is established based on the proportionate distribution of true positives and false negatives in the whole dataset. The F1-score may be computed by obtaining the weighted mean of accuracy and recall.

5. Contribution

i SMOTE techniques and hyper-parameter adjustment to address class disparity and improve disease prediction.
ii LIME is an Explainable AI algorithm developed to provide physicians with a comprehensive understanding of disease factors, including feature importance, visualization, and decision-making importance.
iii An explainable AI technique is used to determine the model's prediction outcome, making it easier to identify key factors.
iv Internal explainability is also utilized to enhance model reliability, balancing accuracy and interpretability. This approach is beneficial for physicians and healthcare professionals.

6. Result and analysis

The experiment of this work is carried out with a data set of nearly 15,300 data records. The relevance of work evaluated on datasets without preprocessing and with preprocessing is explained in the following section.

6.1. Anemia Classification without Preprocessing

The original dataset without preprocessing fed into the models gives 100% accuracy for Logistic Regression, XGBoost, and LGM Boost as shown in Table 1. Decision tree and Random forest have an accuracy of 93% and 98% respectively. 100% accuracy reveals the presence of overfitting issues in those models. Class label imbalance might have led to this issue and this work performs preprocessing on the dataset to overcome it.

Table 1. Classification Results using dataset without and with Preprocessing

Comparison	**Results without preprocessing**				**Results with preprocessing**			
ML Models	**Precision**	**Recall**	**F1-Score**	**Accuracy**	**Precision**	**Recall**	**F1-Score**	**Accuracy**
Logistic reg.	100	100	100	100	73.6	74.2	73.8	74
Decision tree	78.6	79.4	79.2	93	79.8	80	79.8	80
XGBoost	100	100	100	100	85.2	85.6	85.4	86
Random forest	97.6	84.6	89.6	98	87	87.2	86.8	87
LGM Boost	100	100	100	100	90.8	90.8	90.6	91

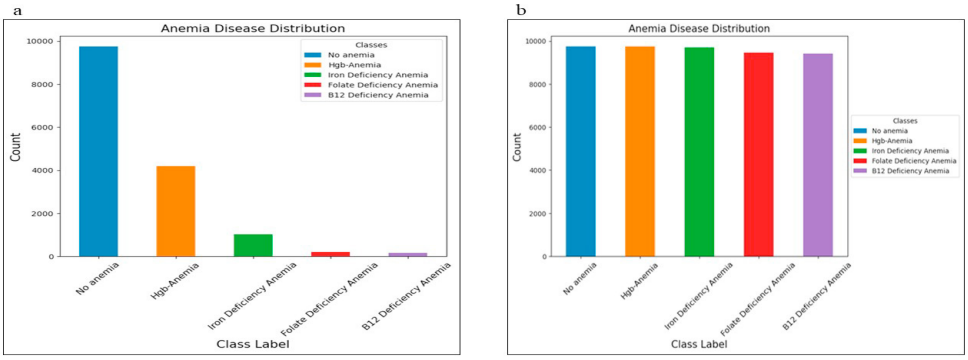


Fig. 2. (a) Unbalanced Data without SMOTE; (b) Balanced Data with SMOTE.

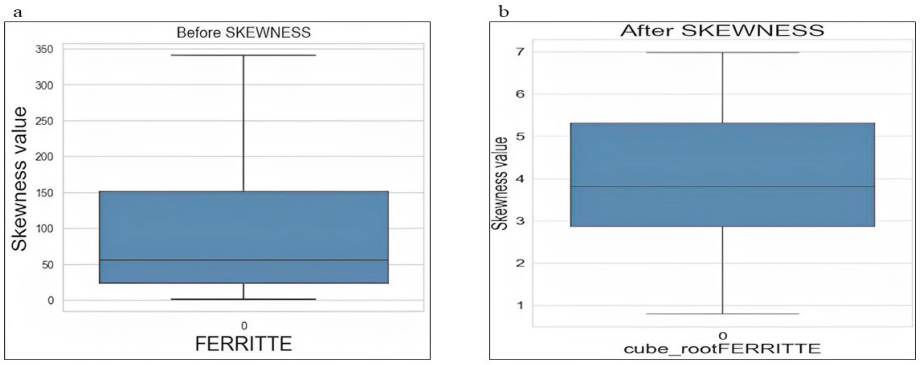


Fig. 3. (a) Before skewness; (b) After skewness.

6.2. Preprocessing in Anemia Dataset

6.2.1. Imbalance class

When one class label has fewer samples than the other, that is known as an imbalance class. In this dataset, the class label no anemia has higher data than all other classes. It is challenging to train the model and generate trustworthy user evaluation predictions because of the class imbalance.

The SMOTE method is used to select instances closest to the feature space by drawing a new instance along the line connecting them. To ensure balanced classes, a representative sample from a minority class is randomly selected, followed by the creation of neighboring instances for the same in a synthetic manner. Fig.2a and Fig. 2b show the imbalance dataset and SMOTE balanced dataset.

Exploratory data analysis techniques are used for outlier removal, data normalization, skewness, kurtosis, covariance, correlation, and variance inflation factor. Fig. 3a with box plot shows the dataset with positive skewness in the variable FERRITTE. It is eliminated through cube root transformation and its result is shown in Fig. 3b. Multicollinearity is identified using the variance inflation factor and eliminated using feature engineering. After the above preprocessing steps overfitting is removed. Anemia dataset after preprocessing is used for training and testing of ML models to produce classification outputs. Comparative results of the model used for classification on non preprocessed dataset exhibits the existence of overfitting problem and it is shown in Fig. 4a. This problem is overcome with preprocessing and the results of those models are shown in Fig 4b. LGM Boost produces the highest score for the evaluation metrics Precision, Recall, and F1 score. Model output is given to LIME explainable AI which provides inferences for the clinical professionals about the main factor causing the prediction output and it is depicted in Fig. 5. The

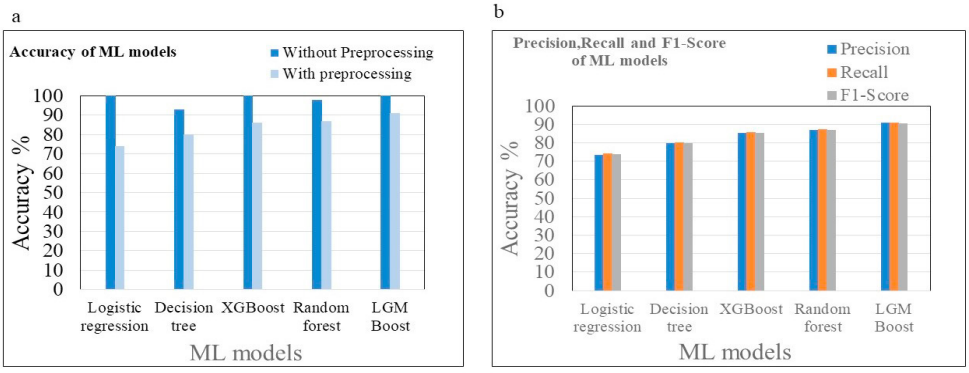


Fig. 4. (a) Accuracy of ML Models with and without Preprocessing; (b) ML models efficiency for preprocessed data.

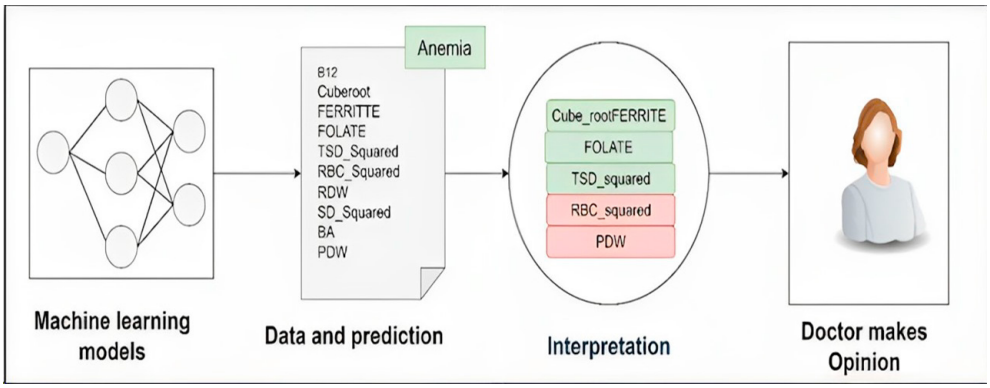


Fig. 5. Anemia Prediction System Using Explainable AI Visualization

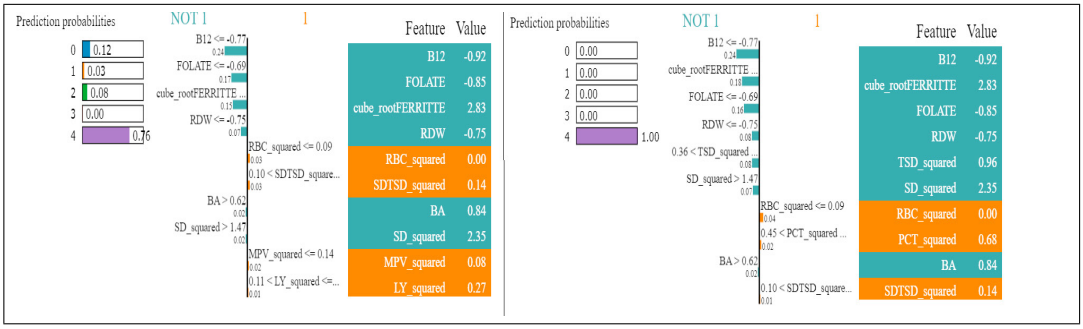


Fig. 6. (a) Explainable AI for Logistic regression with interpretation ; (b) Explainable AI for Random forest with interpretation.

explanation for LIME in explainable AI for the 9th instance in the dataset is shown in Figures 6a,6b,7a,7b and 8. It confirms that the class label's prediction probabilities are 4 in all ML models with the use of Explainable AI. Most machine learning algorithms use a black box approach, whereas Explainable AI uses a white box method to explain why the machine learning system predicts in this way. Explainable AI shows the interpretation regarding the attributes that have more impact in selecting the specific class label. In short, it demonstrates the main causes for that predicted anemia type. It may help the doctor in providing proper treatment based on this information.

Anemia prediction with Explainable AI results reveals that variables `SDTSD_squared`, `LY_squared`, `TSD_squared`, `MPV_squared`,EO are listed in one or two model inferences, and `RBC_Squared` and PWD are listed in 4 ML models. Inference made with LGMBoost model which has the highest accuracy included the above 2 variables `RBC_squared`

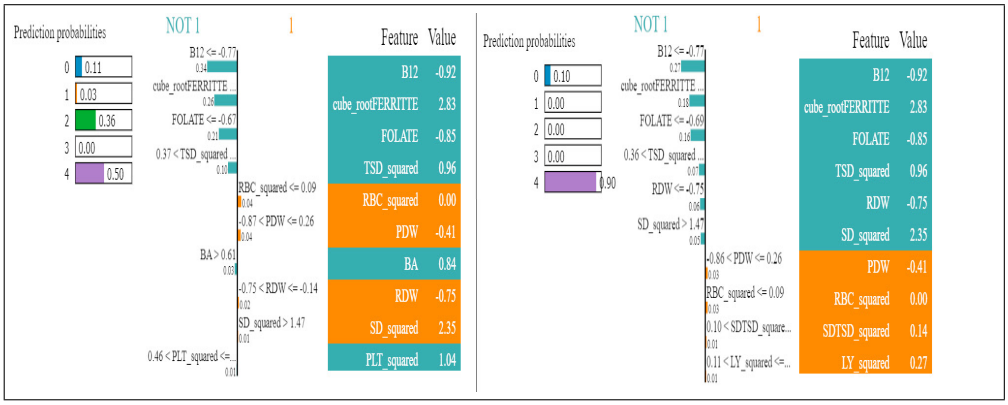


Fig. 7. (a) Explainable AI for Decision tree with interpretation ; (b) Fig.8.Explainable AI for XGBoost with interpretation.

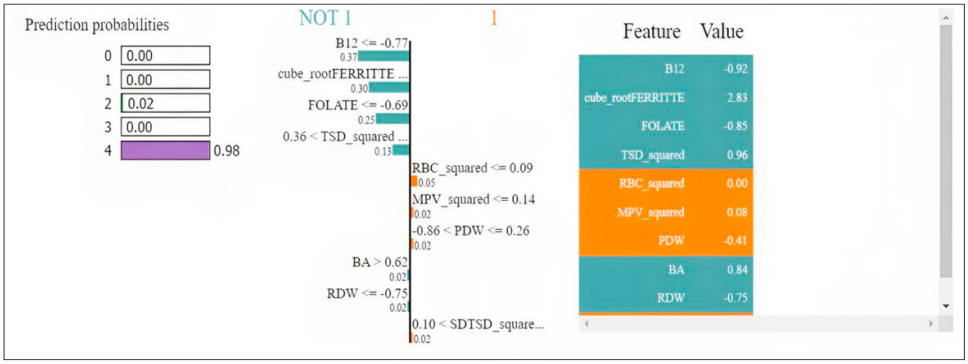


Fig. 8. Explainable AI for LGBoost with interpretation

and PWD. So, clinical professionals can decide their treatment based on those 2 variables, so that anemia can be cured easily as their treatment process will be more specific and accurate.

7. Conclusion

Anemia, a common and potentially life-threatening illness, can be identified and predicted using machine learning algorithms. These models can assess patient data, medical history, demographic information, and behavioral variables to identify those at risk but successful implementation requires careful data quality analysis and model interpretability. Continuous validation and development are necessary to ensure accuracy and usefulness in clinical settings. The use of machine learning and explainable AI in anemia prediction is a significant step forward in healthcare, potentially improving early diagnosis, and patient outcomes, and reducing the burden of this common illness. The dataset is fine-tuned with EDA and five machine-learning models are used to predict the type of anemia. LGMBoost predicts anemia with the highest accuracy and the inference of the variables predicted by this model with explainable AI were listed in 4 out of 5 models used for prediction.More research on this subject would be required since the scope of this study is limited to the development and deployment of a mobile application to increase its utilization. In order to enhance accessibility and usefulness, future research endeavors may include developing a mobile application that integrates the recommended models and is suitable for rural populations.

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